

TEZPUR UNIVERSITY

Autumn Semester End Term Examination, 2021

CO 311: SOFTWAREENGINEERING

Full marks: 60

Time: 3 hours

Figures in the right hand side indicate marks for the question

1. Fill the gaps:

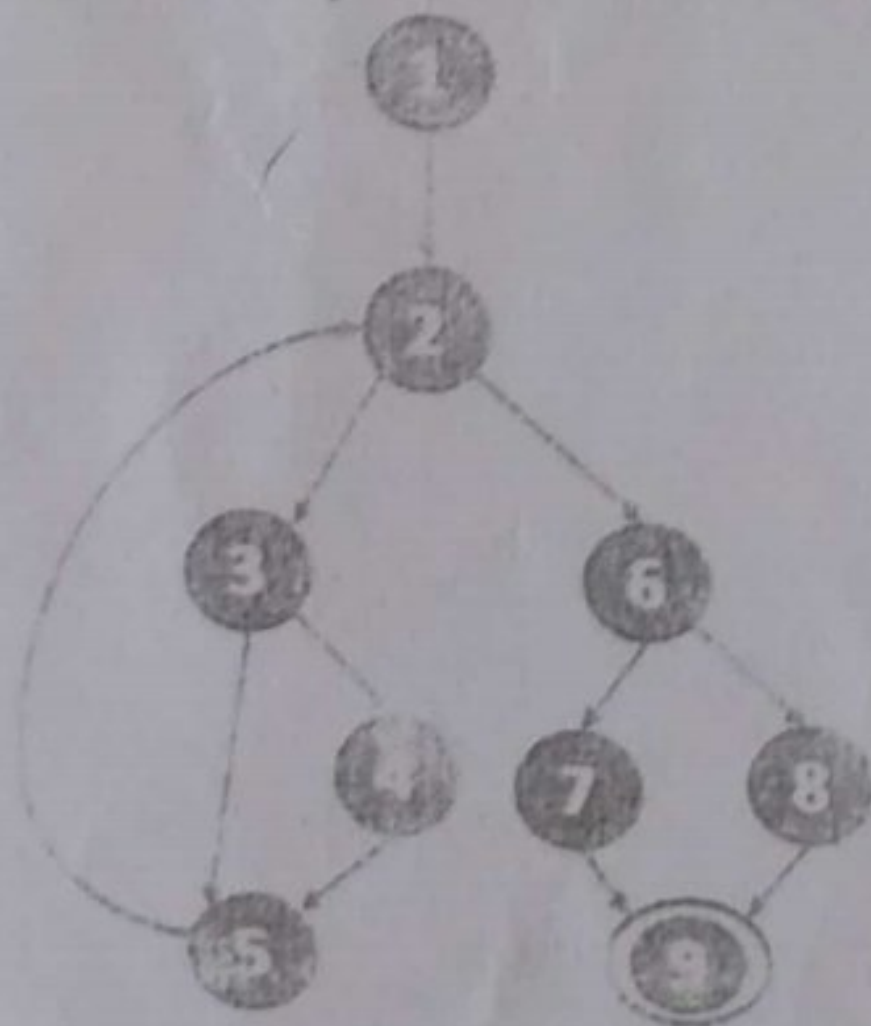
1x10

- The software life cycle model that gives emphasis on risk is _____.
 - The Agile manifesto is based on 4 values and 12 principles.
 - A module having _____ cohesion & _____ coupling is said to be functionally independent of other modules.
 - If two modules share data through some global data items, then they are called to be _____ coupled.
 - If two modules share code, e.g. a branch from one module into another module, then they are called to be _____ coupled.
 - Maintainability, portability, usability etc represented in SRS are coming under _____ requirements.
 - Scrum framework is _____ to understand but _____ to master.
- Describe how mutation testing is suitable for improving initial test suite? 5
 - Write short notes on (i) boundary value analysis (ii) decision Table. 2x5
 - Compare and contrast the lines of code (LOC) and function point (FP) metrics. 5
 - There is a program to checks if an input number between 10 and 100 is a prime number or not. Design test cases for the program using equivalence class partitioning based black box testing approach. 5
 - Compare and contrast hardware and software reliability. 5
 - Size of a *semi detached* type software product has been estimated to be 32,000 lines of source code and average salary of software engineers be Rs. 50,000/- per month. Using *basic* COCOMO model, determine the effort required to develop the software product, the nominal development time. 5
- Answer ANY TWO of the following 2x5
 - Considering the predicate $p = a \vee b$, show that the clause coverage does not subsumes predicate coverage.
 - Define *active clause*. Given the predicate $p = a \vee b$, find out separate set of test cases for clause a and b to be active clauses. Note: Use truth table of the predicate $p = a \vee b$.
 - Explain why parameterized macros are not recommended as per the C language coding standard?

9. Answer ANY ONE of the following

5

- Find Prime paths in the following CFG.



- Find CFG for the following program.

```

#include <stdio.h>
int main() {
    float height;
    printf("Input height(in centimeters): ");
    scanf("%f", &height);
    if (height < 150.0)
        printf("Short. \n");
    else if ((height >= 150.0) && (height < 165.0))
        printf("Average. \n");
    else if ((height >= 165.0) && (height <= 195.0))
        printf("Tall. \n");
    else
        printf("Abnormal. \n");
    return (0);
}

```